

WEEK 1 · DAY 5

IAM & Account Security Setup

BaseStack Academy · AWS Cloud Accelerator

Cloud Foundations

Today's Agenda

IAM & Account Security Setup

1 Overview — Why This Matters

FOUNDATION

2 Key Concepts — Definitions & Vocabulary

CORE KNOWLEDGE

3 Deep Dive — Key Concepts Extended

DEEPER LEARNING

4 How It Works — Step by Step

MECHANICS

5 Technical Deep Dive

ADVANCED DETAIL

6 Architecture & Visual Model

VISUAL

7 Real-World Nigerian Scenario

CONTEXT

8 Common Mistakes & Exam Traps

EXAM PREP

Overview — Why This Matters

The AWS Management Console is your primary visual interface for creating and managing all AWS services. While experienced cloud engineers often use CLI or Infrastructure as Code, learning through the console (ClickOps) gives you the spatial understanding of how services connect and what options are available.

Every configuration you make in the console builds the visual muscle memory you need to understand how AWS services connect, making it much easier when you eventually learn to do it via code.

Billing Alerts are not optional. Cloud bills can surprise even experienced engineers. The moment you start creating resources in AWS, the meter is running. Setting up billing alerts before any lab work is a professional standard — like checking your phone balance before making a call. You need to know your limits so you don't accidentally drain your account!"

WHY NOW?

Cloud skills are the #1 most in-demand tech capability globally. Understanding this topic is essential for both the SAA-C03 exam and real cloud engineering work.



Beginner Note

Do not worry if this feels new — every expert was once a complete beginner. Take notes and ask questions freely.



Key Concepts — Know These Definitions

📖 Study tip: Say each definition aloud — spoken repetition builds memory 3× faster than reading alone.

AWS Management Console

Web browser interface at console.aws.amazon.com — the command centre for all AWS services

AWS Region Selector

Always verify your Region (top-right corner) before creating any resource — resources are Region-specific

AWS Budgets

Set custom cost and usage budgets with automatic email alerts when thresholds are approached or exceeded

CloudWatch Billing Alarms

Specific alarm triggered when estimated monthly charges exceed a dollar threshold you set

AWS Cost Explorer

Visualise, understand, and forecast AWS spending patterns with granular filtering by service and account

Key Concepts (Continued)

Concepts Summary

All key concepts for this lesson have been covered. Use this slide to review and quiz yourself.

✓ AWS Management Console

✓ AWS Region Selector

✓ AWS Budgets

✓ CloudWatch Billing Alarms

✓ AWS Cost Explorer

⚙️ How It Works — Step by Step

1

When you log in to the AWS Console, you land on the personalised home dashboard showing recently visited services, account health alerts, and cost summaries. The global search bar at the top lets you find any of AWS's 200+ services by name or keyword. The Region selector in the top-right corner is critical — most services are regional, but remember: IAM and Billing are global.

2

To set up a CloudWatch Billing Alarm: Switch to us-east-1 Region (billing data is only in this Region) > Navigate to CloudWatch > Alarms > Billing > Create Alarm > Set Metric: Estimated Charges > Greater Than \$5 > Create SNS topic > Add your email > Confirm the subscription link AWS sends you. This \$5 alarm is your financial safety net against forgotten, running lab instances.

3

AWS Budgets is more powerful than CloudWatch alarms — you can set budgets for actual spend, forecasted spend, or Reserved Instance utilisation, and get alerted at 50%, 80%, and 100% thresholds automatically.

Technical Deep Dive — IAM Core Concepts

IAM Identities

Root User

The email address used to create the AWS account. Has unlimited power — never use for daily tasks.

IAM Users

Individual accounts for people or apps. Each has unique credentials and permissions.

IAM Groups

Collections of users. Attach a policy to the group instead of each user. Example: “Developers” group.

IAM Roles

Temporary permissions assumed by services or users. EC2 instances use roles — not user credentials.

IAM Policies

What is a Policy?

A JSON document that defines what actions are allowed or denied on which AWS resources.

Managed Policies

Pre-built by AWS (e.g. AmazonS3ReadOnlyAccess) or custom-built by you. Reusable across identities.

Policy Structure

Effect: Allow / Deny

Action: s3:GetObject, ec2:StartInstances

Resource: specific ARN or * (all)

Least Privilege Principle

Grant only the minimum permissions needed to perform a task. Never grant * on all resources by default.

MFA & Security

What is MFA?

Multi-Factor Authentication adds a second verification step beyond username and password.

Enable MFA immediately on:

- ✓ Root account (non-negotiable)
- ✓ All IAM users with console access
- ✓ Any user with admin permissions

IAM Account Checklist

- Lock root account — never share
- Create individual IAM users
- Assign permissions via groups
- Use roles for AWS service access
- Review IAM Security Score regularly

Architecture & Visual Model — IAM Permission Flow

AWS ACCOUNT BOUNDARY



Root Account

Enable MFA — Never use for daily tasks



IAM Users

Individual identities



IAM Groups

Collections of users



IAM Roles

Temp permissions for services



IAM Policies (JSON: Effect + Action + Resource)



S3



EC2



RDS



Lambda

AWS SERVICES (what IAM controls access to)

KEY PRINCIPLE

- **Root Account** — God-mode access, secure with MFA and lock away
- **IAM Users/Groups** — Who is authenticating
- **IAM Roles** — Temporary assumed permissions for services
- **IAM Policies** — The JSON document that defines what is allowed



Real-World Nigerian Scenario

David is a bootcamp student who forgot to terminate his EC2 instance after a lab exercise. Without billing alerts, he discovered a \$47 charge when his monthly AWS invoice arrived. With a properly configured \$5 billing alert, he would have received an email within 2-3 days of the instance running, giving him plenty of time to stop it before the bill grew. This single habit — setting billing alerts before every lab — saves students hundreds of dollars and eliminates one of the most common fears for beginners: unexpected cloud bills.

Discussion Question

Think about a Nigerian business you know. How would this AWS service change how they operate? What problem would it solve?

⚠️ Common Mistakes & Exam Traps

- These are the most common exam errors — every one costs marks. Memorise them.



Billing and cost data is ONLY available in the US East (N. Virginia) / us-east-1 Region. If you create a CloudWatch billing alarm in any other Region, you will see no data. Always switch to us-east-1 first.



AWS Budgets can send up to 5 alerts for free per month. CloudWatch Billing Alarms incur standard CloudWatch alarm charges after the free tier. For learning, either works — Budgets is more feature-rich.



The Free Tier Usage Alert under AWS Billing settings is a separate, built-in alert that warns when you approach Free Tier limits. Enable this in addition to your manual billing alarm.



This lab creates ZERO billable resources. CloudWatch Billing Alarms are free within the Free Tier. IAM users are always free. MFA setup is free. NO teardown required — keep everything you created. This setup protects your account for the entire duration of your cloud learning journey. Set it once, keep it forever.



Question

A Solutions Architect at a Lagos fintech company is designing the AWS infrastructure for a new payment platform. They need to implement best practices related to IAM & Account Security Setup. Which of the following statements is CORRECT?

A

You should enable MFA on the AWS root account immediately after account creation and avoid using the root account for day-to-day tasks.

✓ CORRECT

B

The root account should be used for everyday AWS tasks because it has the highest permission level and is the most efficient option.

C

IAM users are free but IAM Groups incur hourly charges based on the number of users in the group.

D

IAM policies must always be attached directly to individual users — attaching them to groups is not supported by AWS.

💡 Key Takeaways — What You Learned Today

IAM & Account Security Setup

✓ Concepts Covered

✓ AWS Management Console

✓ AWS Region Selector

✓ AWS Budgets

✓ CloudWatch Billing Alarms

✓ AWS Cost Explorer

📁 Before Next Class

1. Review the concepts above without notes

2. Complete the hands-on console lab

3. Post your progress on LinkedIn / X

4. Review the exam traps from the previous slide

5. Check the Further Resources list



Further Resources & Reading

Go deeper with these official AWS resources:

1. IAM User Guide (Official Documentation)

docs.aws.amazon.com/IAM/latest/UserGuide/

2. IAM Best Practices

docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html

3. Setting Up CloudWatch Billing Alarms

docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/monitor_estimated_charges_with_cloudwatch.html

4. AWS Budgets Getting Started

docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html

5. AWS Free Tier Overview

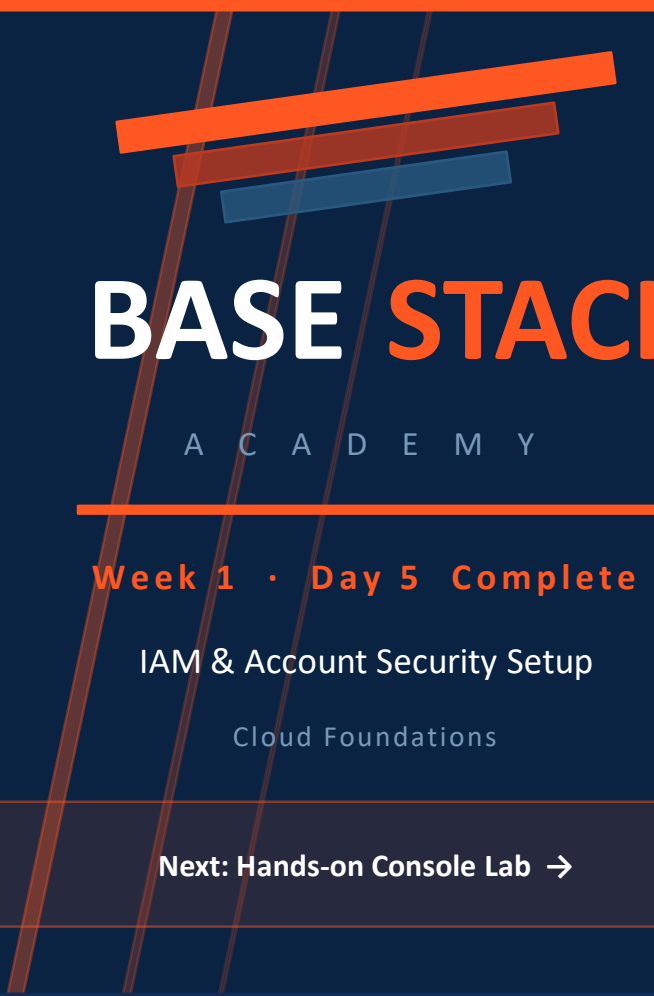
aws.amazon.com/free/

6. SAA-C03 Exam Guide (Pearson VUE)

d1.awsstatic.com/training-and-certification/docs-sa-assoc/AWS-Certified-Solutions-Architect-Associate_Exam-Guide.pdf



Homework: Read at least ONE link above before the next class. Post what you learned on the WhatsApp group.



BASE STACK

A C A D E M Y

Week 1 · Day 5 Complete

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Next: Hands-on Console Lab →